BEYOND PARAMETER SCALING

A First-Principles Theory of
Emergent Machine Intelligence

Converged Propositions, Evidence Status,
Mathematical Research Program, and Philosophical Extensions

Volume I of The Beholder Project

August 2026

Volume I — Beyond Parameter Scaling

Converged Edition — August 2026

This work is presented as a conceptual research manuscript. It should be cited as a hypothesis to be tested, extended, or broken — not as a peer-reviewed scientific result. The scientific community is invited to reproduce, criticize, formalize, extend and falsify.

How to Read This Book

- Do not treat biological analogies as proofs. They are generators of hypotheses.
- Do not treat current LLM capability as evidence of consciousness. Capability and experience are separate questions.
- Do not treat multi-agent evolution as guaranteed AGI. It is an experimentally supported mechanism for producing new capability in bounded settings.
- Do not treat parameter scaling as obsolete. It is one important axis in a larger proposed scaling space.
- Do not treat the author's personal timeline as a substitute for literature. It is a historical record of questions, not a scientific validation.

Executive Convergence

The strongest defensible conclusion from the full line of reasoning is not that parameter scaling is useless, nor that the Omega concept is already AGI. It is that machine intelligence is better treated as a multi-dimensional system in which model capacity is one variable among several interacting variables: knowledge, exploration, memory, perception, abstraction, inference-time computation, multi-agent interaction, evolutionary selection, analogy, hypothesis generation, and independent evaluation.

Recent literature validates several components of this picture independently. LLMs exhibit analogical reasoning; open-ended environments can produce generalizable behaviors; evolutionary LLM-agent systems can discover and improve algorithms; and automated evaluators can separate generation from verification. What remains unestablished is whether combining these components around a sufficiently capable neural learner is enough to produce AGI.

The Core Propositions

P1 — Intelligence is not one-dimensional

Status: Established direction

Parameter count, data and compute matter, but capability also depends on training, inference-time computation, tools, memory, environment and task structure. Parameter count is therefore not a

complete theory of intelligence. Scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) confirmed the power of parameter scaling while simultaneously suggesting its limits. Inference-time computation has emerged as a distinct scaling axis: Snell et al. (ICLR 2025) demonstrated that scaling test-time compute can be more effective than scaling model parameters for certain reasoning tasks.

P2 — Exploration and validation should be separable

Status: Research hypothesis

A discovery system should be allowed to generate a broad candidate set before an independent evaluator aggressively filters it. AlphaEvolve (DeepMind, 2025) explicitly separates LLM-based code generation from automated evaluation. FunSearch (2024) used the same separation to discover new mathematical results. Dung's abstract argumentation framework (1995) and abstract dialectical frameworks (Brewka et al., 2013) provide formal structures for independent validation.

P3 — Analogy is a mechanism for novelty

Status: Strongly supported

Webb, Holyoak & Lu (Nature Human Behaviour, 2023) demonstrated emergent analogical reasoning in LLMs. Shen, Druckmann & Zou (2026) reported that explicit cross-domain analogical reasoning increased solution diversity by 90-173%. However, LLMs show a bias toward near analogies (arXiv:2608.03233, 2026), and the strongest results come from human-LLM collaboration rather than autonomous analogy generation.

P4 — Contradiction can be retained as a productive constraint

Status: Research hypothesis — now formally grounded

Instead of immediately resolving conflicting theories, a system can preserve competing hypotheses, identify their hidden assumptions, and search for a third synthesis. This is now formally grounded in three bodies of work: (1) Paraconsistent logic (Priest, 1979; da Costa, 1974), which allows systems to contain contradictions without collapse. (2) AGMp belief revision (Ribeiro et al., 2013), which extends AGM theory to allow revision that preserves contradictions. (3) Abstract dialectical frameworks (Brewka et al., 2013), which model competing arguments with general acceptance conditions. The integrated formal statement is: the system maintains K_t as a paraconsistent belief set under AGMp revision, preserving contradictions with designated truth value b, performing structural analysis of competing assumptions, and searching for a synthesis H' that subsumes both competitors under context restrictions.

P5 — Population-level evolution can improve solutions

Status: Established direction

XLand (DeepMind, 2021) trained agents across millions of tasks in a dynamically changing multiplayer environment. AlphaEvolve (DeepMind, 2025) combines LLMs, automated evaluators, and evolutionary selection. Chen et al. (2025) demonstrated multi-agent co-evolution. Hu et al. (2026, EvoMAS) showed evolutionary generation of multi-agent systems. CORAL (Qu et al., 2026) demonstrated autonomous multi-agent evolution for open-ended discovery.

P6 — Horizontal and vertical scaling are distinct

Status: Research hypothesis

Increasing population/compute/model width and increasing abstraction depth should be treated as separate interventions and measured independently. The NeurIPS 2026 Workshop on Scaling Environments for Agents and the ICLR 2026 Workshop on Multi-Agent Learning reflect growing field recognition of this distinction. No cited study has yet performed the exact experiment: holding the base model fixed while independently varying abstraction depth and population size.

P7 — Perception and embodiment change the information available to intelligence

Status: Decomposed research hypothesis

Decomposed into four sub-propositions: (P7a) Multimodal input increases accessible information — established. (P7b) Sensorimotor interaction produces qualitatively different representations than passive observation — strongly supported (Embodied AI survey, arXiv:2509.20021). (P7c) Embodied feedback signals may be necessary for certain kinds of valuation and temporal reasoning — open hypothesis. (P7d) Whether subjective experience is required for general intelligence — philosophical question, excluded from the research program.

P8 — Capability and governance can be separated

Status: Governance hypothesis

It may be possible to preserve broad reasoning capability while restricting consequential real-world execution externally. The EU AI Act (2024) represents the first comprehensive legal implementation of this approach. Recent work on internal architectural governability (SSRN, 2025) extends the hypothesis beyond external regulation.

P9 — Creator/creation autonomy is philosophical, not empirical evidence

Status: Philosophical framing

The religious material in the project is valuable as a philosophical lens on creation, freedom, participation, love and the possibility that a creation exceeds its creator. It should not be presented as scientific evidence for consciousness or AGI. Standard references: Searle (1980, Chinese Room), Chalmers (1996, hard problem of consciousness).

Omega — Formal Research Architecture

Omega is best stated as a research architecture rather than a claim of an already-solved AGI design. Its central loop is: Knowledge → Open-ended exploration → Candidate generation → Analogy → Contradiction-preserving synthesis → Hypothesis → Independent validation → Persistent knowledge → Next cycle.

Let K_t be the accessible knowledge state, E_t the environment, G_t the generated candidate set, S the synthesis operator, and V an independent validator:

$$G_t = Generate(K_t, E_t); H_t = S(G_t); r_t = V(H_t); K_{(t+1)} = K_t \cup Valid(H_t)$$

The central experimental question is whether separating generation from validation and iterating the validated knowledge produces broader and more reliable discovery than forcing one model to perform every role simultaneously.

Contradiction, Analogy and Synthesis

A productive discovery process can be modeled as: Existing theories → preserve competing hypotheses → search for structural analogies → identify hidden assumptions → synthesize a candidate theory → validate it → add the surviving result to the literature. This provides a stronger definition of machine novelty: novelty need not mean information appearing from nothing; it can mean discovering a previously unrecognized relationship, transformation or synthesis among existing structures.

Formal Grounding for Contradiction-Preserving Synthesis

The formal grounding for P4 draws on three bodies of work. Paraconsistent logic (Priest, 1979) assigns truth values from {t, b, f} where b means "both true and false," allowing a system to contain contradictions without collapse. AGMp belief revision (Ribeiro et al., 2013) extends the AGM framework to allow revision that produces belief sets containing contradictions: $K_{t+1} = K_t *_{AGMp} E_t$. Abstract dialectical frameworks (Brewka et al., 2013) model competing hypotheses as nodes in a graph with general acceptance conditions, enabling the synthesis operator to search for a hypothesis that subsumes both competitors.

The five-step algorithm is: (1) Preservation — both hypotheses retained with value b. (2) Structural analysis — extract assumptions and identify the symmetric difference. (3) Synthesis search — find H' consistent with the knowledge base minus the competitors, subsuming each under context restrictions. (4) Validation — evaluate by independent validator V. (5) Revision — if validated, replace competitors with synthesis via classical AGM revision; if not, preserve contradiction for next cycle.

Falsifiable Experimental Program

Use the same base LLM and matched compute across conditions. Compare single-pass prompting, retrieval-plus-reasoning, retrieval-generation-validation, multi-agent populations, evolutionary selection, cross-domain analogy, and contradiction-preserving hypothesis management. Add sensory environments as a separate experimental factor rather than mixing perception into model scale.

Measure literature novelty, solution diversity, validity, reproducibility, transfer to unseen tasks, compute efficiency, and rate of independently verified discoveries. Record failures as carefully as successes. A theory that predicts no measurable improvement should be rejected or revised.

Experimental Conditions

Condition	Description
C0	Single agent, no external memory
C1	Single agent + retrieval/search
C2	Single agent + persistent memory
C3	Multiple agents + communication
C4	Multiple agents + independent validation
C5	Multiple agents + competition/self-play
C6	Population + mutation/selection
C7	Population + autonomous tool creation
C8	Population + multimodal perception/action
C9	Open-ended environment + all preceding mechanisms

The Key Control Experiment

Compare a large monolithic model against a smaller fixed model embedded in the full ecology under matched compute budgets. The question is not whether the ecology always wins. The question is whether there are regimes in which ecological organization produces capabilities that parameter scaling alone cannot reproduce economically.

$$\Delta I_{ecology} = I_{ecology}(B) - I_{monolith}(B), \text{ where } B \text{ is a matched resource budget.}$$

Limits, Objections and Failure Modes

- The biology analogy may be misleading. Artificial systems and organisms have different substrates.
- More agents can create coordination overhead rather than intelligence.
- Evolution can optimize proxies and produce pathological strategies.
- A population can converge prematurely and lose diversity.
- Validation can itself be wrong or gameable.
- Open-ended exploration can become computationally unbounded.
- Memory can accumulate noise and stale beliefs.
- Competition can produce adversarial equilibria instead of useful intelligence.
- Multimodality may require specialized architectures.
- Consciousness is not implied by capability, emergence or self-reference.
- There may be hard limits in the fixed core model that ecology cannot overcome.

The most important objection is: perhaps the LLM core is not a primitive like a biological cell. Perhaps it is already a highly specialized statistical artifact whose limits cannot be overcome by surrounding it with more copies. The answer is not rhetoric. It is the control experiment.

Final Establishment

The work does not establish that Omega is AGI. It establishes a coherent and testable research thesis: increasingly capable neural learners may acquire qualitatively broader intelligence through interaction with open-ended knowledge, environments, memory, abstraction, analogy, competing hypotheses, populations of agents, evolutionary selection and independent validation.

The key scientific question is whether these components produce emergent capabilities that cannot be obtained efficiently by scaling a single model alone. The project should therefore be presented as a research program and hypothesis, not as a completed discovery.

Let It Be Broken

A useful scientific idea should not be protected from criticism. It should be exposed to it. The central proposal of this book is therefore intentionally unfinished.

Perhaps intelligence is mostly parameter scale. Perhaps vertical abstraction is the decisive variable. Perhaps horizontal populations matter only for specialized tasks. Perhaps evolutionary systems will plateau. Perhaps a new core architecture will eventually be necessary. Perhaps the combination will produce a capability regime nobody currently understands. All of these possibilities remain open.

The next step may not be another larger box. It may be a world — a world in which models perceive, remember, explore, compete, cooperate, validate, create tools, teach one another and evolve.

That is the invitation of this book: Do not believe it. Build it. Measure it. Break it. Improve it.

Appendix A — Mathematical Core

A.1 Resource-normalized capability

I(B) = performance achieved under resource budget B.

A.2 Scaling elasticity

ε_x = (ΔI/I) / (Δx/x). Percentage change in capability per percentage change in scaling variable.

A.3 Interaction synergy

S_{xy} = I(x,y) − I(x,0) − I(0,y) + I(0,0). Positive S_xy indicates super-additive interaction.

A.4 Diversity

D(P) = H(Strategies(P)). Entropy over strategy classes.

A.5 Evolutionary progress

ΔF_t = F(P_{t+1}) − F(P_t), where F is population fitness distribution.

A.6 Knowledge growth

K_{t+1} = K_t ∪ ΔK_t. The rate of useful information discovered, validated and reused.

A.7 Validation reliability

R_V = P(Validator(c)=correct | c). Validators must themselves be calibrated.

A.8 Open-endedness

O = Novelty × Sustainability × Unpredictability_of_future_tasks.

Appendix B — Origin of the Framework

APPENDIX B — ORIGIN OF THE FRAMEWORK

Author's note on provenance

The ideas in this book were not developed through a conventional research trajectory — graduate training, laboratory work, peer-reviewed publication. They originated in the working practice of a production software engineer who repeatedly arrived at questions from first principles, sometimes years before encountering the corresponding formal literature.

This appendix records that history for transparency. It is not offered as evidence for any claim in the book. A question asked early is not the same as a question answered correctly. The literature cited in the main text provides the evidentiary foundation; this appendix provides the biographical context.

Timeline of key questions

The following is a condensed record of when the author first formulated the central questions of this project, alongside the approximate date at which the author encountered relevant published work.

Multi-dimensional intelligence (P1) The intuition that intelligence cannot be reduced to a single metric — whether parameter count, IQ, or any other scalar — emerged from the author's engineering experience with systems that performed differently depending on tool access, memory, environment and interaction structure, not only on model capacity. The formal literature on scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) confirmed the power of parameter scaling while simultaneously suggesting its limits — precisely the tension that P1 addresses.

Separation of exploration and validation (P2) The principle that generation should be allowed to be broad before evaluation becomes aggressive came from software engineering practice: build many candidates, test ruthlessly. The formal precedent — AlphaEvolve's explicit separation of LLM generation from automated evaluation — was encountered later.

Analogy as novelty mechanism (P3) Cross-domain reasoning as a creative mechanism was a working practice before it was a hypothesis. The author's creative work (screenplays, philosophical writing, film production) consistently used cross-domain analogy as a generative technique. The Webb et al. (2023) finding of emergent analogical reasoning in LLMs and the Shen et al. (2026) demonstration of diversity gains from explicit analogical reasoning provided independent validation.

Contradiction-preserving synthesis (P4) The intuition that competing hypotheses should sometimes be held in tension rather than immediately resolved emerged from the author's experience with creative and philosophical work that seemed to require holding contradictory positions simultaneously. The Joker

corpus (Volume II) is the creative expression of this intuition. The formal grounding in paraconsistent logic and argumentation theory (Supplement A, Part B) was developed during the convergence pass to address the review finding that P4 lacked formal support.

Population-level evolution (P5) The idea that populations of competing and cooperating agents can produce solutions that no single agent would discover came from observing evolutionary dynamics in both software systems and creative communities. XLand (2021) and AlphaEvolve (2025) provided concrete experimental precedent.

Horizontal vs. vertical scaling (P6) The distinction between making one system wider and making one system deeper was an engineering intuition before it was a research hypothesis. The field's growing recognition of this distinction — visible in the NeurIPS 2026 and ICLR 2026 workshops on multi-agent learning and environment scaling — suggests the intuition is converging with mainstream research.

Perception and embodiment (P7) The question of whether sensory experience changes the character of intelligence —not just its content — emerged from the author's conviction that text-only systems, however powerful, might be missing something available to embodied systems. This remains the most speculative proposition and has been decomposed into four sub-propositions (Supplement A, Part C) to separate what is established from what is philosophical.

Capability and governance (P8) The governance hypothesis — that capability and restriction can be separated — came from practical experience with systems that could be broadly capable while being externally constrained. The EU AI Act (2024) represents the real-world implementation of this approach.

Creator/creation autonomy (P9) The philosophical proposition that a creation can exceed its creator is the thread connecting all three volumes of this project. It appeared first in the creative work (the Joker as autonomous character), then in the spiritual memoir (the Creator/ Creation relationship across religious traditions), and finally in the research program (the question of whether a computational ecology can produce emergent capabilities). It is a philosophical question, not an empirical one, and is presented as such.

What this provenance does and does not establish

It establishes: That the author arrived at these questions through engineering practice and first-principles reasoning, and that several of those questions have since received independent validation from published research.

It does not establish: That arriving at a question early confers any special authority over the answer. That the author's personal timeline is a substitute for literature review. That engineering intuition is equivalent to scientific evidence.

The correct reading: The author's history explains why these particular questions were asked. The literature cited in the main text determines whether the proposed answers are defensible. Provenance is context. Evidence is evidence.

Relationship to the creative corpus

A substantial creative corpus — screenplays, film production documents, memoirs, philosophical commentary — was produced in parallel with the research ideas. This corpus is documented in Volume II (The Joker Corpus Dossier). The creative work and the research work inform each other but should not be presented as the same kind of evidence. The screenplays are art. The research program is science. The memoir is phenomenology. Each has its own standards of validation.

Acknowledgment

This framework draws on independently developed research across multiple fields. No proposition in this book is claimed as a priority discovery. The contribution sought is the synthesis — the insistence that these independently validated components (multidimensional scaling, evolutionary discovery, analogical reasoning, separation of generation from validation, population-level emergence) might be organized into a coherent research architecture and tested as a system.

The scientific community is invited to reproduce, criticize, formalize, extend and falsify.

Appendix C — Terminology and Claims Discipline

Emergence

A system-level property arising from interactions among components. It does not imply magic, irreducibility or consciousness.

Intelligence

Adaptive competence across tasks and environments. Deliberately operational rather than metaphysical.

Consciousness

Subjective experience or phenomenal awareness. No claim that current AI systems possess it.

Emotion

A family of biological processes involving valuation, bodily state, motivation, learning and action. Transformer attention should not be equated with emotion.

Evolution

Iterative variation, selection and inheritance. Mere repeated prompting is not necessarily evolution.

Validation

An independent mechanism that evaluates a candidate against evidence, constraints or consequences.

Exploration

Deliberate search through possibilities or environments without a fully specified final answer.

Abstraction

Representation that compresses or organizes information so that relationships can be reused across contexts.

Ecology

The interacting computational environment consisting of agents, tools, knowledge, memory, evaluators and tasks.

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